

Diffusion and Score Matching

Chapter 1: Deep Generative Modeling Basics (English-main)

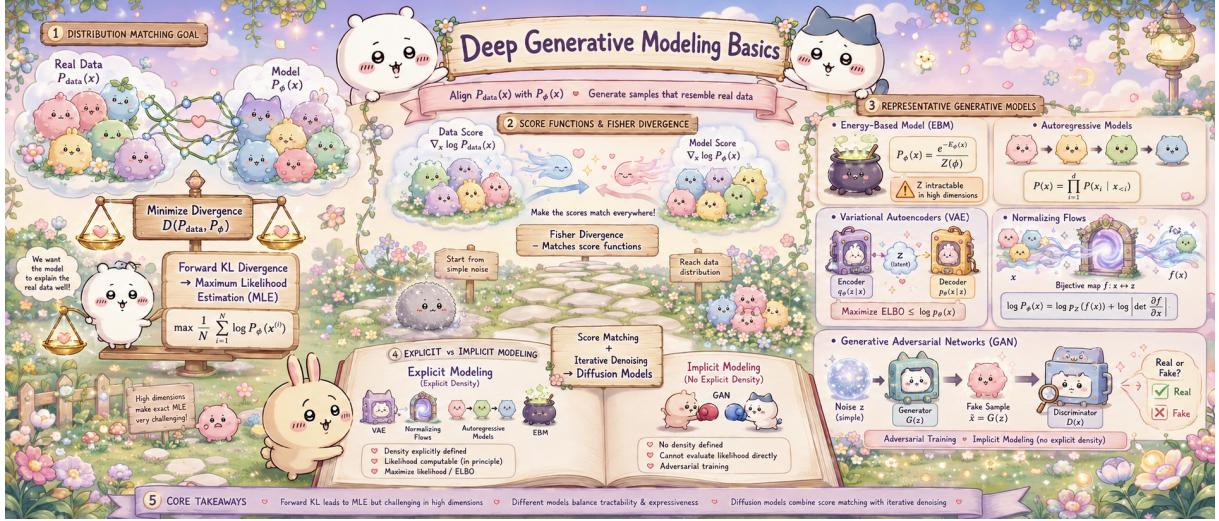
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Chapter 1 · Diffusion and Score Matching

Setup and Goal

Divergences and Training

Taxonomy and Summary



1 Deep Generative Modeling

1.1 Problem Setup

Input: A collection of real-world samples drawn from an unknown data distribution $P_{\text{data}}(\mathbf{x})$.

Output: A neural-network parameterized model distribution $P_{\phi}(\mathbf{x})$.

Goal:

- (1) Generate samples that resemble real data;
- (2) Enable controllable generation.

1.2 Mathematical Formulation

Let the training set be

$$\mathcal{D} = \{\mathbf{x}^{(i)}\}_{i=1}^N, \quad \mathbf{x}^{(i)} \stackrel{\text{i.i.d.}}{\sim} P_{\text{data}}.$$

We seek parameters ϕ^* such that

$$P_{\phi^*}(\mathbf{x}) \approx P_{\text{data}}(\mathbf{x}).$$

Training is framed as minimizing a discrepancy between distributions:

$$\phi^* = \arg \min_{\phi} D(P_{\text{data}}, P_{\phi}).$$

2 Divergences and Training Objectives

2.1 KL Divergence (Forward KL)

Definition 1: Kullback–Leibler Divergence

$$D_{\text{KL}}(P_{\text{data}} \parallel P_{\phi}) \triangleq \int P_{\text{data}}(\mathbf{x}) \log \frac{P_{\text{data}}(\mathbf{x})}{P_{\phi}(\mathbf{x})} d\mathbf{x} = \mathbb{E}_{\mathbf{x} \sim P_{\text{data}}} \left[\log \frac{P_{\text{data}}(\mathbf{x})}{P_{\phi}(\mathbf{x})} \right].$$

Expanding yields

$$D_{\text{KL}}(P_{\text{data}} \parallel P_{\phi}) = \mathbb{E}_{\mathbf{x} \sim P_{\text{data}}} [\log P_{\text{data}}(\mathbf{x}) - \log P_{\phi}(\mathbf{x})] = -\mathbb{E}_{\mathbf{x} \sim P_{\text{data}}} [\log P_{\phi}(\mathbf{x})] - H(P_{\text{data}}),$$

where

$$H(P_{\text{data}}) \triangleq -\mathbb{E}_{\mathbf{x} \sim P_{\text{data}}} [\log P_{\text{data}}(\mathbf{x})]$$

is the entropy of the data distribution and does not depend on ϕ . Therefore:

Theorem 2: Forward KL $\iff$ Maximum Likelihood Estimation

$$\min_{\phi} D_{\text{KL}}(P_{\text{data}} \parallel P_{\phi}) \iff \max_{\phi} \mathbb{E}_{\mathbf{x} \sim P_{\text{data}}} [\log P_{\phi}(\mathbf{x})].$$

With finite data, this corresponds to the empirical objective

$$\max_{\phi} \frac{1}{N} \sum_{i=1}^N \log P_{\phi}(\mathbf{x}^{(i)}).$$

2.2 Fisher Divergence**Definition 3: Fisher Divergence**

$$D_F(P \parallel Q) = \mathbb{E}_{\mathbf{x} \sim P} [\|\nabla_{\mathbf{x}} \log P(\mathbf{x}) - \nabla_{\mathbf{x}} \log Q(\mathbf{x})\|^2].$$

This divergence is always nonnegative: $D_F(P \parallel Q) \geq 0$. Under suitable regularity conditions,

$$D_F(P \parallel Q) = 0 \iff \nabla_{\mathbf{x}} \log P(\mathbf{x}) = \nabla_{\mathbf{x}} \log Q(\mathbf{x}) \quad (P\text{-a.s.}).$$

In diffusion/score matching settings, one typically sets $P = P_{\text{data}}$ and $Q = P_{\phi}$, so training matches score functions.

2.3 f -Divergence Family**Definition 4: f -Divergence**

For convex $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ with $f(1) = 0$:

$$D_f(P \parallel Q) = \int Q(\mathbf{x}) f\left(\frac{P(\mathbf{x})}{Q(\mathbf{x})}\right) d\mathbf{x}.$$

Common members:

- **Jensen–Shannon Divergence**

$$f_{\text{JS}}(u) = u \log u - (u + 1) \log \frac{u + 1}{2},$$

$$D_{\text{JS}}(P \parallel Q) = \frac{1}{2} D_{\text{KL}}(P \parallel M) + \frac{1}{2} D_{\text{KL}}(Q \parallel M), \quad M = \frac{P + Q}{2}.$$

- **Total Variation Distance**

$$f_{\text{TV}}(u) = \frac{1}{2}|u - 1|,$$

$$D_{\text{TV}}(P, Q) = \frac{1}{2} \int |P(\mathbf{x}) - Q(\mathbf{x})| d\mathbf{x} = \sup_A |P(A) - Q(A)|.$$

2.4 Challenges in Explicit Density Modeling

A valid density $P_\phi(\mathbf{x})$ must satisfy:

1. Non-negativity: $P_\phi(\mathbf{x}) \geq 0$;
2. Normalization: $\int P_\phi(\mathbf{x}) d\mathbf{x} = 1$.

A common construction is the Energy-Based Model (EBM), written in two steps:

1. Define an energy network $E_\phi : \mathbb{R}^D \rightarrow \mathbb{R}$ and obtain an unnormalized density

$$\tilde{P}_\phi(\mathbf{x}) = \exp(E_\phi(\mathbf{x}));$$

2. Normalize it with the partition function

$$P_\phi(\mathbf{x}) = \frac{\exp(E_\phi(\mathbf{x}))}{Z(\phi)}, \quad Z(\phi) = \int \exp(E_\phi(\mathbf{x})) d\mathbf{x}.$$

In practice, the partition function $Z(\phi)$ is usually intractable in high dimensions.

3 Representative Generative Models

1. Energy-Based Models (EBM)

Define an energy function $E_\phi : \mathbb{R}^D \rightarrow \mathbb{R}$ via a neural network (using the **negative-energy** convention):

$$P_\phi(\mathbf{x}) \triangleq \frac{1}{Z(\phi)} \exp(-E_\phi(\mathbf{x})), \quad Z(\phi) = \int \exp(-E_\phi(\mathbf{x})) d\mathbf{x}.$$

The partition function $Z(\phi)$ is generally intractable in high dimensions.

2. Autoregressive Models

Factorize the joint distribution by the chain rule:

$$P(\mathbf{x}) = \prod_{i=1}^D P_\phi(x_i | x_{<i}), \quad x_{<i} \triangleq (x_1, x_2, \dots, x_{i-1}).$$

Each conditional $P_\phi(x_i | x_{<i})$ is parameterized by a neural network (e.g. a Transformer).
Training objective: minimize the negative log-likelihood

$$\min_{\phi} -\frac{1}{N} \sum_{n=1}^N \sum_{i=1}^D \log P_\phi(x_i^{(n)} | x_{<i}^{(n)}).$$

3. Variational Autoencoders (VAE)

Introduce a latent variable z to capture hidden structure in data:

- Encoder: $q_\phi(z | \mathbf{x})$
- Decoder: $p_\theta(\mathbf{x} | z)$

Training objective: maximize the Evidence Lower Bound (ELBO)

$$\mathcal{L}_{\text{ELBO}}(\theta, \phi; \mathbf{x}) = \mathbb{E}_{z \sim q_\phi(z|\mathbf{x})} [\log p_\theta(\mathbf{x} | z)] - D_{\text{KL}}(q_\phi(z | \mathbf{x}) \parallel p(z)) \leq \log p_\theta(\mathbf{x}).$$

4. Normalizing Flows

Learn a **bijective mapping** f between data space $\mathbf{x}$ and latent space z :

$$\mathbf{x} \xrightarrow{f} z \xrightarrow{f^{-1}} \mathbf{x}'.$$

By the change-of-variables formula:

$$\log P_\phi(\mathbf{x}) = \log p_Z(f(\mathbf{x})) + \log \left| \det \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} \right|.$$

With a simple prior p_Z (e.g. standard Gaussian), log-likelihood is computed exactly.

5. Generative Adversarial Networks (GAN)

GAN trains two networks adversarially:

- **Discriminator** D : distinguishes real from generated samples

$$\mathbf{x} \longrightarrow D(\mathbf{x}) \longrightarrow 0/1$$

- **Generator** G : maps prior noise to fake samples

$$z \sim p_{\text{prior}}(z) \longrightarrow G(z) \longrightarrow \mathbf{x}'$$

Standard minimax objective:

$$\min_G \max_D \mathbb{E}_{\mathbf{x} \sim P_{\text{data}}} [\log D(\mathbf{x})] + \mathbb{E}_{z \sim p_{\text{prior}}(z)} [\log(1 - D(G(z)))].$$

Key point: GAN does **not** define a density $P(\mathbf{x})$ —it is a canonical example of implicit modeling.

3.1 Explicit vs. Implicit Modeling

	Explicit Modeling	Implicit Modeling
Density $P(\mathbf{x})$	Explicitly defined (or approximated)	Not defined
Likelihood	Computable (exactly or approximately)	Cannot be directly evaluated
Training	Maximize likelihood / ELBO	Adversarial training
Sampling	Via model distribution	$\mathbf{x} = G_\phi(z)$, $z \sim p_{\text{prior}}$
Examples	VAE, Normalizing Flows, AR, EBM	GAN

4 Summary

- The core objective is to align P_ϕ with P_{data} ;
- Forward KL leads to MLE but can be computationally challenging in high dimensions;
- Different model classes balance tractability and expressiveness differently;
- Diffusion models can be viewed as score matching plus iterative denoising, to be covered next.